

AI Refinement of a Heath–Brown Type Zero-Free Region for the Riemann Zeta Function

From the 4.8913 Baseline to a 4.761794 Internal Candidate

<https://github.com/BlinkDL/VibeMath>

I would say this is 99% AI, although I provided some inputs and guidance.

August 2026

Abstract

This manuscript develops a certificate-based refinement of the Bellotti–Trudgian–Yang zero-free-region argument. The original body reaches the internal candidate 4.8913; the first supplement explains the subsequent route to 4.8038. The present revision adds a second supplement describing the later chain to

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.761794 \log t} \right).$$

The additional improvements are structural rather than a single parameter change. A positive-definite product replaces the termwise strip-boundary majorant; the principal Laplace expression is evaluated on its exact bootstrap trajectory; the central digamma integral and endpoint W_0 term collapse to one Bose-kernel integral; the full smoothing weight is retained inside every prime-power orbit; a degree-eight SOS factor gives global half-line positivity; the Heath–Brown parameter is reoptimized; and a third-harmonic rigidity inequality recovers a certified large-prime tail. All reported values remain computer-assisted internal candidates. Exact integer, rational, Sturm, Bernstein, and SOS computations are separated from directed interval calculations, and an independent Arb or MPFI replay and a direct audit of the external Yang inputs are still required.

1 Introduction

For $s = \sigma + it$ with $\sigma > 1$, the logarithmic derivative

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}$$

turns nonnegative prime-power coefficients into a zero detector. Classical arguments combine this positivity with a nonnegative trigonometric polynomial. Stechkin’s use of the functional equation and Heath–Brown’s compactly supported smoothing function make this detector substantially stronger. Bellotti, Trudgian and Yang [1] combine these two ideas and obtain

$$\sigma > 1 - \frac{1}{4.896 \log t}, \quad t \geq 3.$$

Their proof has three numerically sensitive components:

- (i) the nonnegative trigonometric polynomial;
- (ii) a lower bound for grouped prime-power orbits;
- (iii) the finite fixed-point ledger after the explicit formula is applied.

The present note modifies all three components, but keeps the external analytic inputs of [1] unchanged: Ford’s estimate for the Laplace transform, the smoothed explicit formula, the zero-counting estimates, the verification of the Riemann hypothesis up to $3 \cdot 10^{12}$, and Yang’s explicit Littlewood region. The genuinely new computations are:

- (1) an exact degree-twelve Fejér square P_{12} ;
- (2) exact prime-power lower bounds for every prime $p < 100$;
- (3) a direct interval treatment of the dominant $m = 0$ Laplace integral;
- (4) a corrected and sharpened remainder ledger;
- (5) the final fixed-point calculation for $R = 4.8913$.

We formulate the main statement in certificate-dependent form.

Theorem 1.1 (Computer-assisted refinement). *Assume the external explicit estimates used in [1], including the verification of the Riemann hypothesis up to $H = 3 \cdot 10^{12}$ and the explicit Littlewood region*

$$\sigma > 1 - \frac{\log \log t}{21.233 \log t}.$$

If the exact rational certificates and directed interval enclosures described in Sections 4–6 are replayed, then

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8913 \log t} \right).$$

Verification status 1.2. The arithmetic certificates use exact integers and rational numbers. The remaining one-dimensional integrals and parameter inequalities use directed interval arithmetic. The note records the full mathematical reduction and the numerical margins, but it is not a proof-assistant formalization. An independent Arb, MPFI, or comparable replay is therefore desirable.

2 The smoothing function and normalization

We retain the smoothing function of [1]. Fix

$$\theta = 1.1338, \quad U = 2\theta \cot \theta,$$

and define, for $0 \leq u \leq U$,

$$\begin{aligned} w(u) = \sec^2 \theta & \left(\sec^2 \theta \left(\theta \cot \theta - \frac{u}{2} \right) \cos(u \tan \theta) + 2\theta \cot \theta - u \right. \\ & \left. + \frac{\sin(2\theta - u \tan \theta)}{\sin 2\theta} - 2 \left(1 + \frac{\sin(\theta - u \tan \theta)}{\sin \theta} \right) \right), \end{aligned} \quad (1)$$

with $w(u) = 0$ outside $[0, U]$. Numerically,

$$U = 1.0592329388789857\dots, \quad w(0) = 5.6727875988616045\dots$$

Let

$$(\kappa_0, \dots, \kappa_6) = \left(1, -\frac{851}{859}, \frac{780}{859}, -\frac{525}{859}, \frac{171}{859}, \frac{28}{859}, -\frac{29}{859} \right),$$

so that

$$\kappa := \sum_{m=0}^6 \kappa_m = \frac{433}{859}.$$

For $\delta_m = (2\sigma - 1)m$, define

$$f(u) = \eta w(\eta u) \sum_{m=0}^6 \kappa_m e^{-\delta_m u}, \quad F(z) = \int_0^\infty e^{-zu} f(u) du.$$

To avoid a harmless normalization ambiguity in displayed formulas, we set

$$f_*(0) := \eta w(0).$$

Thus the literal value of the function in (2) is $f(0) = \kappa f_*(0)$, while the explicit-formula ledger is most naturally written in units of $f_*(0)$.

Two elementary positivity facts are useful. First, if

$$p(x) = \sum_{m=0}^6 \kappa_m x^m,$$

then

$$859p(x) = \sum_{j=0}^6 b_j \binom{6}{j} x^j (1-x)^{6-j}, \quad (2)$$

where

$$(b_0, \dots, b_6) = \left(859, \frac{4303}{6}, \frac{1882}{3}, \frac{2253}{4}, \frac{7651}{15}, 469, 433 \right).$$

Hence

$$p(x) \geq \frac{433}{859} = \kappa \quad (0 \leq x \leq 1).$$

Second, w is decreasing. Indeed, after putting $v = u \tan \theta$, direct simplification gives

$$w'(u) = \sec^2 \theta \sin v \left[\tan \theta + \sec^2 \theta \left(\frac{v}{2} - \theta \right) - \tan \frac{v}{2} \right].$$

Convexity of $\tan$ on $[0, \theta]$ makes the bracket nonpositive, so $w'(u) \leq 0$.

For the final parameter range, directed interval subdivision verifies

$$f(u) \geq \kappa \eta w(0) \quad (0 \leq u \leq 59). \quad (3)$$

3 The degree-twelve Fejér square

Define

$$\begin{aligned} Q_{12}(z) = & 32635 - 54033z - 17946z^2 + 174507z^3 + 66931z^4 - 374536z^5 \\ & - 168219z^6 + 1652944z^7 + 4446322z^8 + 6169255z^9 + 5432436z^{10} \\ & + 3003044z^{11} + 837164z^{12}. \end{aligned} \quad (4)$$

Its squared Euclidean norm is

$$\mathcal{N} := \sum_{j=0}^{12} q_j^2 = 100000002139070.$$

We take

$$P_{12}(x) = \frac{|Q_{12}(e^{ix})|^2}{\mathcal{N}} = 1 + \sum_{k=1}^{12} a_k \cos(kx). \quad (5)$$

Nonnegativity is automatic. The exact numerators $\mathcal{N}a_k$ are

$$(173779234139896, 111942599517052, 49767351006488, 12146409523234, \\ 9367972, 9065744, 1076858103392, 588712579182, 2346122, 654528, \\ 105539717056, 54641694280).$$

Every entry is positive. The principal parameters are

$$a_1 = 1.737792304226366158 \dots, \quad (6)$$

$$a := \sum_{k=1}^{12} a_k = 3.494613602397228815 \dots, \quad (7)$$

$$P_{12}(0) = 1 + a = 4.494613602397228815 \dots. \quad (8)$$

The polynomial was obtained by a constrained numerical search, followed by integer rationalization. Only the displayed integer vector is used in the proof.

4 Exact prime-power orbit certificates

For a prime p and an integer $M \geq 1$, set

$$G_{p,M}(\sigma, x) = \sum_{m=1}^M p^{-m\sigma} P_{12}(mx).$$

Since $P_{12} \geq 0$,

$$\frac{\partial}{\partial \sigma} G_{p,M}(\sigma, x) = -(\log p) \sum_{m=1}^M m p^{-m\sigma} P_{12}(mx) \leq 0. \quad (9)$$

Thus $G_{p,M}(\sigma, x) \geq G_{p,M}(1, x)$ whenever $\sigma \leq 1$.

For strict compatibility with the support range in (3), use

$$M_p = \min \left(15, \left\lfloor \frac{59}{\log p} \right\rfloor \right).$$

The infinite orbit is explicit. Writing $y = \cos x$ and denoting Chebyshev polynomials by T_k , one has

$$G_{p,\infty}(1, x) = \frac{1}{p-1} + \sum_{k=1}^{12} a_k \frac{pT_k(y) - 1}{p^2 - 2pT_k(y) + 1}. \quad (10)$$

Moreover,

$$0 \leq G_{p,\infty}(1, x) - G_{p,M_p}(1, x) \leq \frac{P_{12}(0)}{(p-1)p^{M_p}}. \quad (11)$$

For each prime $p < 100$, subtracting a rational candidate b_p , applying (11), and multiplying by the twelve positive denominators in (10) gives an integer polynomial $R_p(y)$ of degree at most 78. On replacing y by $2u - 1$, R_p is certified positive on $[0, 1]$ by exact dyadic Bernstein subdivision. The resulting lower bounds are shown in Table 1.

The logarithms are also bounded from below by rational arithmetic, using the positive truncation

$$\log r = 2 \sum_{j=0}^{J-1} \frac{u^{2j+1}}{2j+1} + \text{positive remainder}, \quad u = \frac{r-1}{r+1},$$

p	b_p	p	b_p
2	0.2383955346	43	0.0000071512
3	0.0518827925	47	0.0000054858
5	0.0075903925	53	0.0000038329
7	0.0021992359	59	0.0000027820
11	0.0004621014	61	0.0000025181
13	0.0002683487	67	0.0000019018
17	0.0001156281	71	0.0000015988
19	0.0000822567	73	0.0000014712
23	0.0000461824	79	0.0000011612
29	0.0000230998	83	0.0000010015
31	0.0000189391	89	0.0000008126
37	0.0000111868	97	0.0000006279
41	0.0000082412		

Table 1: Certified lower bounds for $G_{p,M_p}(1, x)$.

after extracting a power of two. This yields

$$\sum_{p < 100} (\log p) b_p > 0.2415849. \quad (12)$$

The high-precision value is

$$0.2415849157806932048498958094 \dots$$

Proposition 4.1 (Improved arithmetic term). *For the smoothing function in Section 2 and the polynomial P_{12} ,*

$$\sum_{n \geq 1} \frac{\Lambda(n)}{n^\sigma} f(\log n) P_{12}(t \log n) > 0.1217767 \eta w(0). \quad (13)$$

Proof. Group the terms with $n = p^m$, use (3), (9), and the orbit certificates:

$$\begin{aligned} \sum_{n \geq 1} \frac{\Lambda(n)}{n^\sigma} f(\log n) P_{12}(t \log n) &\geq \kappa \eta w(0) \sum_{p < 100} (\log p) G_{p,M_p}(1, t \log p) \\ &> \frac{433}{859} \cdot 0.2415849 \eta w(0) \\ &> 0.1217767 \eta w(0). \end{aligned}$$

□

The corresponding constant in [1] is 0.1186. Thus the arithmetic part alone improves by approximately 2.68%.

5 Recomputed analytic ledger

We now record only the changes needed in the explicit-formula calculation. The notation follows [1]. Put

$$\mu = \frac{1 - \sigma}{\eta}, \quad T_j(\sigma) = \sum_{m=0}^6 \frac{\kappa_m}{(1 + (2\sigma - 1)m)^j}.$$

Clearing denominators shows algebraically that

$$T'_1(\sigma) > 0, \quad T'_2(\sigma) > 0 \quad (\sigma \geq 1/2).$$

No floating-point root isolation is required.

The reciprocal expansion is

$$W\left(\frac{\delta_m}{\eta} + x\right) = \frac{w(0)}{\delta_m}\eta - \frac{xw(0)}{\delta_m^2}\eta^2 + E_m(\eta). \quad (14)$$

For $x = -\mu$ and $x = 1 - \mu$, the signs in the quadratic terms must be read directly from (14). Uniform estimates use $|x| \leq 1$, and Ford's parameter is scaled by η^{-1} . With these choices, the combined $m \geq 1$ terms and the $F(\sigma - \eta)$ term satisfy the following safe bound throughout the final parameter box:

$$\begin{aligned} a_1 F(\sigma - 1 + \eta) + a_1 F(\sigma - \eta) - F(\sigma - 1) \\ \geq I_Q(\mu) + 3.9090\eta + 20.189\eta^2 - 170\eta^3, \end{aligned} \quad (15)$$

where

$$I_Q(\mu) = \int_0^U e^{\mu u} (a_1 e^{-u} - 1) w(u) du. \quad (16)$$

The direct treatment of I_Q is important. The cubic Taylor minorant used for a coarse uniform estimate loses more than the final fixed-point margin. On the actual interval needed in Section 6, directed interval integration gives

$$I_Q(\mu) > 0.99348, \quad 0.136 < I'_Q(\mu) < 0.138. \quad (17)$$

The gamma-factor estimate gives a gain greater than $1.5630\eta w(0)$. Combining this with Proposition 4.1, the explicit-formula errors

$$53a\eta^2 + (2353 + 1601a)\eta^3,$$

and (15), one obtains

$$\frac{a\kappa w(0)}{2} \eta \log t \geq I_Q(\mu) + C_Q(\eta) - 10^{-7}, \quad (18)$$

where the safe cubic polynomial

$$C_Q(\eta) = 13.4663\eta - 165.026\eta^2 - 8117.877\eta^3 \quad (19)$$

may be used. Finally,

$$\frac{a\kappa w(0)}{2} = 4.9964370814383408 \dots < 4.99644. \quad (20)$$

For completeness, a separate interval audit of the remaining numerical lemmas gives the following safe statements in the present parameter range:

- the $t = 0$ integral occurring in the gamma error is less than 34.652; the two contributions still sum to less than $309\eta^3 < 623\eta^3$;
- for $t \geq H$, the error remains bounded by $14\eta^2 + 424\eta^3$;
- the far-zero loss remains below 10^{-8} per detector and below 10^{-7} after the trigonometric weights are summed;
- the terms $F(s_k - 1)$ with $k \geq 1$ are negligible compared with the published $10^{-10}\eta$ allowance.

6 The fixed-point calculation

Set

$$\begin{aligned} R &= 4.8913, & A_0 &= \frac{1}{R} = 0.2044446261730010\dots, \\ H &= 3 \cdot 10^{12}, & T_0 &= 10^{10}, & K &= 12. \end{aligned}$$

Define

$$\begin{aligned} \eta_0 &= \frac{A_0}{\log H} = 0.007116158542429939\dots, \\ \sigma_0 &= 1 - \frac{A_0}{\log(KH + T_0)} = 0.993450398220328925\dots \end{aligned}$$

In particular,

$$\frac{2\sigma_0 - 1}{\eta_0} = 138.6844869400089\dots > 138,$$

so the auxiliary Ford and nonnegativity estimates remain inside their certified ranges.

The explicit Littlewood region takes over once

$$\log t \geq L_* := \exp(21.233A_0) = 76.78219271137694\dots$$

It is therefore enough to treat

$$H \leq t \leq e^{L_*}.$$

Under the usual contradiction hypothesis for the iteration lemma,

$$\eta > \frac{1}{6 \log t}.$$

Moreover, throughout this interval one may use

$$\mu \geq 1 - \frac{c_K}{\log t}, \quad c_K = \log \left(12 + \frac{1}{300} \right) + 10^{-6} = 2.485185388992674\dots$$

At the worst endpoint $\log t = L_*$ this gives

$$\mu_* = 0.9676333105211719\dots, \tag{21}$$

$$\eta_* = \frac{1}{6L_*} = 0.002170642186439818\dots \tag{22}$$

The function

$$\mathcal{B}(L) = I_Q \left(1 - \frac{c_K}{L} \right) + C_Q \left(\frac{1}{6L} \right)$$

is decreasing on $[\log H, L_*]$. Indeed,

$$L^2 \mathcal{B}'(L) = c_K I'_Q(\mu) - \frac{1}{6} C'_Q(\eta),$$

and the interval bounds in (17), together with the explicit polynomial (19), make the right-hand side strictly negative. Thus the endpoint is the only value that must be checked.

At that endpoint,

$$I_Q(\mu_*) > 0.99348, \quad C_Q(\eta_*) > 0.02836.$$

Using (18) and (20),

$$\begin{aligned} \eta \log t &> \frac{0.99348 + 0.02836 - 10^{-7}}{4.99644} \\ &> 0.2045135 \\ &> 0.2044446261730010 = A_0. \end{aligned}$$

This contradicts the assumption that a zero satisfies

$$(1 - \beta) \log t < A_0 + 10^{-100}.$$

Hence the iteration advances by 10^{-100} until the value A_0 is reached. The verified-zero computation below H and Yang's Littlewood region above e^{L^*} complete the proof of Theorem 1.1, subject to the certificate status stated after the theorem.

7 Supplement: from 4.8913 to 4.8038

The preceding sections reproduce the original 4.8913 baseline. The present supplement records the later refinements. It is important to distinguish logical dependence from methodological continuity.

Remark 7.1 (Additional external inputs). The 4.8038 candidate is not obtained from the 4.8913 region alone. The later bootstrap also uses the external classical seed

$$\sigma > 1 - \frac{1}{4.862 \log t}$$

and the Littlewood region with constant 19.62. These constants are recorded by Lee [7] as consequences of Yang's work. Thus the supplement refines the computational method of the 4.8913 manuscript, but it imports a stronger starting region.

Theorem 7.2 (Supplementary certificate statement). *Assume the external analytic and verified-zero inputs used in [1], together with the classical seed 4.862 and the Littlewood constant 19.62 recorded in [7]. Assume also that the exact rational, Sturm, Bernstein, finite-moment, and directed-interval certificates described in this supplement are replayed. Then the supplemented computation gives*

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.8038 \log t} \right).$$

Verification status 7.3. The statement is an internally closed computer-assisted candidate. The finite algebraic certificates are exact, while the one-dimensional integral and fixed-point bounds use directed interval arithmetic. Neither an independent Arb/MPFI replay nor a third-party line-by-line review has yet been completed. The Yang thesis itself was not independently audited here; the constants 4.862 and 19.62 are used through the explicit report in [7].

7.1 The correct non-circular finite bootstrap

Let R_s be a zero-free constant already proved and let $R_t < R_s$ be the stronger target. It is convenient to work with widths

$$A = \frac{1}{R_s}, \quad B = \frac{1}{R_t}, \quad B > A.$$

If a hypothetical zero is written as

$$\rho_0 = \beta_0 + it, \quad \eta = 1 - \beta_0, \quad L = \log t,$$

then the seed region and the target contradiction hypothesis give

$$\eta L \geq A, \quad \eta L < B. \tag{23}$$

The auxiliary line must be defined from the already known width A :

$$\sigma = 1 - \frac{A}{\log(Ke^L + T_0)}. \tag{24}$$

It is not legitimate to substitute the target width B into (24) before B has been established. Put

$$c_H = \log \left(K + \frac{T_0}{H} \right).$$

From (23) and $\log(Ke^L + T_0) \leq L + c_H$ one obtains

$$\eta \geq \frac{A}{L}, \quad \mu := \frac{1 - \sigma}{\eta} > \frac{A}{B} \frac{L}{L + c_H}. \quad (25)$$

The factor A/B in the second inequality is essential.

Let $D_Q = \alpha \kappa w(0)/2$ be the normalizing denominator and let $C_{1,Q}$ and $C_{2,Q}$ denote the certified lower-bound functions after the explicit formula is assembled. A finite step $A \rightarrow B$ is proved once

$$\mathcal{L}_{A,B}(L) := \frac{C_{1,Q}\left(\frac{A}{B} \frac{L}{L+c_H}\right) + C_{2,Q}(A/L) - 10^{-7}}{D_Q} > B \quad (26)$$

throughout

$$\log H \leq L \leq \exp(19.62B).$$

Above the right endpoint the Littlewood region is already stronger than the target classical region. In every step used below, interval differentiation shows that the left-hand side of (26) is decreasing, so only the right endpoint needs to be checked.

This finite-step formulation replaces a formally circular fixed-point substitution by a genuine induction in which every target uses only a previously certified seed.

7.2 A second Fejér square and direct finite prime-power certificates

The first large improvement after the 4.8913 baseline comes from replacing (4) by

$$\begin{aligned} Q_{12}^{(2)}(z) = & 2973020 - 5101549z - 1556727z^2 + 17311464z^3 + 5669545z^4 \\ & - 38313778z^5 - 13979425z^6 + 174340757z^7 + 455260827z^8 \\ & + 620102516z^9 + 535373941z^{10} + 288437326z^{11} + 77213430z^{12}. \end{aligned} \quad (27)$$

Its squared norm is

$$\mathcal{N}_2 = 1000000000489220451,$$

and

$$P_{12}^{(2)}(x) = \frac{|Q_{12}^{(2)}(e^{ix})|^2}{\mathcal{N}_2} = 1 + \sum_{k=1}^{12} a_k^{(2)} \cos(kx). \quad (28)$$

The exact numerators $\mathcal{N}_2 a_k^{(2)}$ are

$$\begin{aligned} & (1736607730982643766, 1116308926917508272, 494498266956404318, \\ & 120058469375169958, 428646472, 314671138, 10351059402375478, \\ & 5575245011467526, 258706458, 101620472, 927243684682900, 459114143317200). \end{aligned}$$

Every entry is positive. In particular,

$$a_1^{(2)} = 1.736607730133059749 \dots, \quad \sum_{k=1}^{12} a_k^{(2)} = 3.484786055872385352 \dots$$

For a prime p and

$$M_p = \min \left(15, \left\lfloor \frac{57.5}{\log p} \right\rfloor \right),$$

consider the finite orbit

$$G_{p,M_p}^{(2)}(x) = \sum_{m=1}^{M_p} p^{-m} P_{12}^{(2)}(mx).$$

Unlike the baseline calculation, no infinite orbit and no crude geometric tail are introduced. After setting $y = \cos x$ and $y = 2u - 1$, clearing the positive factor $\mathcal{N}_2 p^{M_p}$ turns $G_{p,M_p}^{(2)}(x) - b_p$ into an integer polynomial in u of degree at most $12M_p \leq 180$. Exact dyadic de Casteljau subdivision proves positivity of its Bernstein coefficients on every terminal subinterval. The result is

$$\boxed{\sum_{p < 100} (\log p) b_p > 0.2439742.} \quad (29)$$

The previous raw floor was 0.2415849, so the gain at this stage is about 0.989%.

For the last pre-Fourier bootstrap vector

$$10^{10}(\kappa_0, \dots, \kappa_6) = (10^{10}, -9986207037, 9793152105, -8692402447, 6137992779, -2719409222, 501106985). \quad (30)$$

we have

$$\kappa = 0.5034233163,$$

and hence the certified arithmetic contribution is

$$\boxed{\kappa \cdot 0.2439742 = 0.12282230085563946 \dots} \quad (31)$$

The corresponding baseline value in Proposition 4.1 was 0.1217767.

7.3 The pre-Fourier stage: reaching 4.8062105

The coefficients κ_m are reoptimized subject to four families of exact constraints:

- (i) Bernstein positivity and monotonicity of $p(x) = \sum \kappa_m x^m$ on $[0, 1]$;
- (ii) global positivity of the Lemma 5 rational function after clearing denominators, certified by a Sturm sequence on $X \geq 0$;
- (iii) coefficientwise negativity of the cleared numerators proving $T'_1(\sigma), T'_2(\sigma) > 0$;
- (iv) directed-interval versions of the smoothing, Ford-remainder, gamma, and fixed-point inequalities.

At the numerical optimum the scaled Lemma 5 function $(1 + X)B_\kappa(X)$ has three nearly equal local minima,

$$X_1 \approx 0.635099, \quad X_2 \approx 15.18344, \quad X_3 \approx 357.68873.$$

The six approximate equations consisting of equal values and vanishing derivatives at these three points explain why the degree-six vector is almost rigid.

Using the seed 4.862, the Littlewood constant 19.62, the polynomial $P_{12}^{(2)}$, and stepwise rationalized κ -vectors gives the following non-circular chain. Thus the new arithmetic polynomial, the exact finite orbits, the actual Ford constants, and the corrected finite-step logic together move the internal candidate from the 4.8913 baseline to 4.8062105.

seed R_s	target R_t	directed-interval margin
4.862	4.8136	$6.17875422 \times 10^{-6}$
4.8136	4.80719	$1.90908374 \times 10^{-7}$
4.80719	4.8064	$3.04840982 \times 10^{-6}$
4.8064	4.80625	$8.64922615 \times 10^{-7}$
4.80625	4.80622	$2.62874828 \times 10^{-7}$
4.80622	4.806218	$3.70844830 \times 10^{-7}$
4.806218	4.806212	$5.20349606 \times 10^{-8}$
4.806212	4.8062105	$1.02037588 \times 10^{-8}$

Table 2: The finite bootstrap before the signed-shift improvement.

7.4 Signed-shift Fourier collapse

The last improvement comes from postponing absolute values in the central explicit-formula remainder. Put

$$d = 2\sigma - 1, \quad r = \frac{d}{\eta}, \quad D = 2\theta \cot \theta.$$

In the notation of the BTY Laplace-transform estimate, write

$$W_0(z) = W(z) - \frac{w(0)}{z}.$$

Define

$$h(x) = \begin{cases} w(x) - w(0), & 0 \leq x \leq D, \\ -w(0), & x > D. \end{cases} \quad (32)$$

Since

$$\frac{w(0)}{z} = \int_0^\infty e^{-zx} w(0) dx,$$

we have the exact representation

$$W_0(z) = \int_0^\infty e^{-zx} h(x) dx. \quad (33)$$

The central error contains the signed combination

$$K_r(v) = \sum_{m=0}^6 \kappa_m W_0\left(r \left(m + \frac{1}{2}\right) + iv\right). \quad (34)$$

Let

$$p(u) = \sum_{m=0}^6 \kappa_m u^m.$$

Substituting (33) into (34) and retaining the signs until after summation gives

$$K_r(v) = \int_0^\infty e^{-ivx} e^{-rx/2} \sum_{m=0}^6 \kappa_m e^{-rmx} h(x) dx \quad (35)$$

$$= \int_0^\infty e^{-ivx} g_r(x) dx, \quad (36)$$

where

$$\boxed{g_r(x) = e^{-rx/2} p(e^{-rx}) h(x)}. \quad (37)$$

Thus seven shifted transforms collapse to one Fourier transform $K_r = \widehat{g}_r$.

Differentiating (37) introduces

$$q(u) = \frac{1}{2}p(u) + up'(u) = \sum_{m=0}^6 \left(m + \frac{1}{2}\right) \kappa_m u^m, \quad (38)$$

and

$$g'_r(x) = e^{-rx/2} [p(e^{-rx})h'(x) - rq(e^{-rx})h(x)]. \quad (39)$$

For the seed vector (30), exact Bernstein coefficients prove

$$p(u) > 0, \quad q(u) > 0 \quad (0 \leq u \leq 1). \quad (40)$$

The same check is repeated for every rationalized vector in the final bootstrap.

7.5 Finite rational moments and the Carlson–Sobolev bound

Let

$$c = -w''(0).$$

Directed interval arithmetic gives

$$c < 51.74, \quad \sup_{0 \leq x \leq D} |w'''(x)| < 144. \quad (41)$$

Since $h(0) = h'(0) = 0$ and $h''(0) = -c$, the certified Taylor bounds are

$$\left| h(x) + \frac{c}{2}x^2 \right| \leq 24x^3, \quad |h'(x) + cx| \leq 72x^2. \quad (42)$$

The extension beyond D is checked at the endpoint and then follows from the polynomial growth of the right-hand sides.

Put $y = rx$. Four quantities control the two Sobolev norms:

$$I_4 = \int_0^\infty e^{-y} p(e^{-y})^2 y^4 dy, \quad (43)$$

$$I_6 = \int_0^\infty e^{-y} p(e^{-y})^2 y^6 dy, \quad (44)$$

$$J = \int_0^\infty e^{-y} \left(\frac{1}{2} q(e^{-y}) y^2 - p(e^{-y}) y \right)^2 dy, \quad (45)$$

$$E = \int_0^\infty e^{-y} \left(\frac{1}{2} p(e^{-y}) y^2 + \frac{1}{6} q(e^{-y}) y^3 \right)^2 dy. \quad (46)$$

They are finite rational sums, not numerical quadratures. For example,

$$I_j = j! \sum_{m,n=0}^6 \frac{\kappa_m \kappa_n}{(m+n+1)^{j+1}}. \quad (47)$$

For the seed vector,

$$I_4 < 22.73, \quad I_6 < 709.61, \quad (48)$$

$$J < 0.474, \quad E < 19.96. \quad (49)$$

Consequently,

$$\|g_r\|_2 \leq r^{-5/2} \left(\frac{c}{2} \sqrt{I_4} + \frac{24}{r} \sqrt{I_6} \right), \quad (50)$$

$$\|g'_r\|_2 \leq r^{-3/2} \left(c\sqrt{J} + \frac{144}{r}\sqrt{E} \right). \quad (51)$$

Replaying the exact moment sums for the stepwise vectors gives, uniformly for $r \geq 136$,

$$\sqrt{A_r B_r} < 71.879405, \quad (52)$$

where A_r, B_r denote the bracketed quantities in (50)–(51).

For the Fourier convention $\widehat{g}(v) = \int_{\mathbb{R}} e^{-ivx} g(x) dx$, Cauchy–Schwarz and Plancherel give, for any $a > 0$,

$$\|\widehat{g}\|_1^2 \leq 2\pi^2 \left(a\|g\|_2^2 + \frac{\|g'\|_2^2}{a} \right).$$

Choosing $a = \|g'\|_2 / \|g\|_2$ yields the Carlson–Sobolev inequality

$$\|\widehat{g}\|_1 \leq 2\pi \sqrt{\|g\|_2 \|g'\|_2}. \quad (53)$$

Combining (52) and (53) gives the safe uniform certificate

$$\boxed{\|K_r\|_1 \leq \frac{2\pi \cdot 72}{r^2} \quad (r \geq 136).} \quad (54)$$

7.6 The improved central explicit-formula error

The symmetric central interval in the BTY error term satisfies

$$\left| \sum_{m=0}^6 \kappa_m E_{2,\text{central}}(s + \delta_m) \right| \leq \frac{\eta \log t}{2\pi} \|K_r\|_1. \quad (55)$$

If the current target window gives $\eta \log t \leq A_{\text{tar}}$, then $r = d/\eta$ and (54) imply

$$\boxed{\left| \sum_m \kappa_m E_{2,\text{central}}(s + \delta_m) \right| \leq \frac{72 A_{\text{tar}}}{d^2} \eta^2.} \quad (56)$$

At the final parameters the coefficient in (56) is approximately

$$15.3961.$$

The previous sharpened ledger, after taking seven absolute values separately, used approximately

$$29.4452.$$

Thus the central quadratic loss is almost halved. Boundary and cubic terms are not reoptimized here; they retain the previously certified conservative bounds.

7.7 The final bootstrap to 4.8038

The pre-Fourier stage reaches 4.8062105. For the signed-shift stage we use the slightly weaker certified seed 4.806212 and the following finite chain. The first row uses (30). The later rows use the rational vectors

$$10^{10} \kappa^{(2)} = (10^{10}, -9991606743, 9851742295, -8933214378, 6558980863, -3049313048, 597531062), \quad (57)$$

$$10^{10} \kappa^{(3)} = (10^{10}, -9991606724, 9851742454, -8933211200, 6558980641, -3049309663, 597526570), \quad (58)$$

$$10^{10} \kappa^{(4)} = (10^{10}, -9991606879, 9851744444, -8933217397, 6558995470, -3049319800, 597527505). \quad (59)$$

seed R_s	target R_t	directed-interval margin
4.806212	4.8043	1.18546×10^{-5}
4.8043	4.8040	1.16384×10^{-5}
4.8040	4.8039	8.18572×10^{-6}
4.8039	4.8038	3.29239×10^{-6}

Table 3: The signed-shift finite bootstrap.

Each vector is rechecked independently for Bernstein positivity of p and q , Lemma 5 Sturm positivity, $T'_1, T'_2 > 0$, Ford remainders, smoothing positivity, and the finite-step criterion (26). This proves the supplementary certificate statement, subject to its stated status.

A target 4.8037 was also tested. The best proof slack found in the current search was about -8×10^{-6} . This is not a no-go theorem, but it shows that the present degree-six κ family, $Q_{12}^{(2)}$, smoothing parameter, and $C = 72$ Fourier certificate are close to a local wall.

8 Second supplement: from 4.8038 to 4.761794

The first supplement ended with the internally certified candidate $R = 4.8038$. The present supplement records the subsequent refinements down to

a robust reporting value

$$\boxed{R = 4.761794}.$$

The logical status is the same as in Theorem 7.2: every finite integer, rational, Sturm, Bernstein, and interval statement described below must be replayed, and the result is not asserted as a published theorem.

Theorem 8.1 (Second supplementary certificate statement). *Assume the external analytic inputs of [1], the classical seed 4.862 and the Littlewood constant 19.62 recorded in [7], and the complete certificate chain of Theorem 7.2. Assume in addition that the positive-definite boundary, exact-trajectory, combined-digamma, fully weighted prime-power, degree-eight SOS, optimized- θ , and large-prime-tail certificates of this section are replayed. Then the supplemented computation gives*

$$\zeta(\sigma + it) \neq 0 \quad \left(t \geq 3, \quad \sigma > 1 - \frac{1}{4.761794 \log t} \right).$$

Verification status 8.2. The strongest thin endpoint found in the same final ledger is approximately 4.7617935, but its interval margin is only about $4.5 \cdot 10^{-9}$. The value 4.761794 is therefore used throughout this manuscript. The result is an internally closed computer-assisted candidate. It has not been replayed in an independent Arb or MPFI implementation, the external Yang inputs have not been rederived here, and a third-party line-by-line review is still missing.

8.1 Overview of the second refinement

The improvement is not caused by one new decimal optimization. It is a sequence of structural replacements:

$$\begin{aligned}
4.8038 & \xrightarrow{\text{sharper signed-shift norm}} 4.80347 \\
& \xrightarrow{\text{positive-definite strip boundary}} 4.77775 \\
& \xrightarrow{\text{exact trajectory in Lemma 14}} 4.77357 \\
& \xrightarrow{\text{exact gamma and combined } E_2} 4.77079 \\
& \xrightarrow{\text{fully weighted prime-power orbits}} 4.76215 \\
& \xrightarrow{\text{degree-eight SOS factor}} 4.7618105 \\
& \xrightarrow{\theta\text{-optimization}} 4.76180 \\
& \xrightarrow{\text{large-prime third-harmonic tail}} 4.761794.
\end{aligned} \tag{60}$$

Every arrow is realized by a finite non-circular bootstrap of the form (26); the auxiliary line is always defined from the already certified seed width.

8.2 A near-optimal signed-shift Fourier certificate

Section 7.4 used

$$\|K_r\|_1 \leq \frac{2\pi \cdot 72}{r^2}.$$

The constant 72 can be reduced by retaining the quadratic and cubic jets of h exactly. Write

$$h(x) = \frac{h_2}{2}x^2 + \frac{h_3}{6}x^3 + \rho_4(x), \quad h_j = h^{(j)}(0),$$

and define

$$S_j(\xi) = \sum_{m=0}^6 \frac{\kappa_m}{(m + \frac{1}{2} + i\xi)^j}.$$

After the scaling $v = r\xi$ one obtains

$$r^3 K_r(r\xi) = h_2 S_3(\xi) + \frac{h_3}{r} S_4(\xi) + \mathcal{R}_r(\xi). \tag{61}$$

For each fixed ξ , the modulus of the first two terms is convex as a function of $1/r$. Hence its supremum on a compact r -interval occurs at an endpoint. Rational midpoint–Lipschitz integration of these endpoint kernels, together with a fourth-order Carlson–Sobolev estimate for $\mathcal{R}_r$, gives the uniform certificate

$$\boxed{\|K_r\|_1 \leq \frac{2\pi \cdot 63.816}{r^2} \quad (r \geq 135.289).} \tag{62}$$

The certified decomposition is

$$C_{\text{main}} < 63.6220, \quad C_{\text{rem}} < 0.1935.$$

Using (62) in the central explicit-formula error replaces the coefficient $72A_{\text{tar}}/d^2$ by $63.816A_{\text{tar}}/d^2$. The resulting finite step is

$$4.8038 \longrightarrow 4.80347,$$

with a directed-interval margin greater than $2.9 \cdot 10^{-7}$.

8.3 An exact positive-definite strip boundary

The numerical wall after 4.80347 came from the auxiliary constant c_0 in the termwise W_0 majorant of Lemma 5. This wall can be removed completely. Let

$$p(x) = \sum_{m=0}^6 \kappa_m x^m, \quad Q_\kappa(x) = (1+x)p(x) = \sum_{m=0}^7 b_m x^m,$$

with

$$10^{12}(\kappa_0, \dots, \kappa_6) = (10^{12}, -999992599356, 997836264493, \\ -956384638133, 755637550446, -381718762624, 84897721337). \quad (63)$$

The Fourier transform of $Q_\kappa(e^{-r|x|})$ is

$$2\pi b_0 \delta_0 + \frac{2}{r} \sum_{m=1}^7 \frac{b_m m}{m^2 + (\xi/r)^2} d\xi.$$

After clearing the positive denominator and writing $Y = (\xi/r)^2$, the density numerator is

$$\mathcal{P}(Y) = 12999997Y^6 + 1260999739Y^5 + 91749275505891Y^4 \\ + 821106817177Y^3 + 7418807583172Y^2 + 29026281915984Y + 38102381432640. \quad (64)$$

All coefficients are positive, so $Q_\kappa(e^{-r|x|})$ is positive definite for every $r > 0$. The even extension $w(|x|)$ is also positive definite, because its Fourier transform is $2 \operatorname{Re} W(i\xi) \geq 0$. Their pointwise product is therefore positive definite. Consequently

$$\operatorname{Re} \sum_{m=0}^7 b_m W(rm + iv) = \int_0^\infty w(x) Q_\kappa(e^{-rx}) \cos(vx) dx \geq 0 \quad (65)$$

for all $r > 0$ and $v \in \mathbb{R}$.

Equation (65) replaces the entire c_0 -Sturm majorant in the first part of Lemma 5. In particular, there is no longer a lower threshold such as $r \geq 138$ coming from that boundary argument. After recomputing the remaining Ford and far-zero terms, one obtains the finite chain

$$4.80347 \rightarrow 4.7812 \rightarrow 4.7783 \rightarrow 4.77783 \rightarrow 4.77778 \rightarrow 4.77776 \rightarrow 4.77775.$$

The smallest margin in this chain is greater than $1.2 \cdot 10^{-7}$.

8.4 The exact trajectory form of the main Laplace expression

The previous Lemma 14 ledger treated

$$\mu = \frac{1-\sigma}{\eta} \quad \text{and} \quad \eta$$

as independent variables in a rectangle. Along an actual finite bootstrap step they lie on a one-dimensional trajectory. Put

$$d = 2\sigma - 1, \quad r = \frac{d}{\eta}, \quad p(z) = \sum_m \kappa_m z^m.$$

The complete shifted expression has the exact integral representation

$$\mathcal{E}(\mu, \eta) := a_1 F(\sigma - 1 + \eta) + a_1 F(\sigma - \eta) - F(\sigma - 1) \\ = \int_0^D w(s) e^{\mu s} p(e^{-rs}) \left(a_1 e^{-s} - 1 + a_1 e^{-s/\eta+s} \right) ds. \quad (66)$$

No Taylor truncation in μ or η is used in (66).

Let

$$z = e^{-rs}, \quad A = a_1 e^{-s} - 1, \quad B = a_1 e^{-s/\eta+s}, \quad C = A + B,$$

and

$$q(z) = \frac{1}{2}p(z) + zp'(z).$$

Differentiation gives

$$\mathcal{E}_\mu = 2 \int_0^D w(s) sq(z) e^{\mu s} C \, ds, \quad (67)$$

$$\mathcal{E}_\eta = \eta^{-2} \int_0^D w(s) s e^{\mu s} (zp'(z)C + p(z)B) \, ds. \quad (68)$$

For the rational SOS vectors used below, exact Bernstein certificates prove positivity on $[0, 1]$ of

$$p, \quad q, \quad L(z) := p(z) + zp'(z) + \frac{p'(z)}{c}, \quad c = \frac{a_1}{a_1 - 1},$$

as well as $-L'$ and $1 + p'$. These polynomial facts make the signs and broad bounds in (67)–(68) effective. On the complete parameter box needed here, directed intervals give representative bounds

$$0.134 < \mathcal{E}_\mu < 0.146, \quad 1.18 < \mathcal{E}_\eta < 16.7.$$

Writing $x = \eta \log t$ and $L = \log t$, the full fixed-point margin is strictly decreasing in both variables. Hence every finite step reduces to one endpoint calculation. This exact-trajectory replacement, together with an SOS positive-definite κ vector, gives

$$4.77775 \longrightarrow 4.77357.$$

8.5 Exact gamma channels and the Bose-kernel collapse of E_2

The gamma term is evaluated channel by channel at its actual trajectory argument

$$1 + \frac{\sigma + (2\sigma - 1)m}{2} = 1 + \frac{(2m + 1)\sigma - m}{2},$$

rather than at a rectangular endpoint. This produces a small but systematic increase in the archimedean contribution.

A larger simplification occurs in the explicit-formula error. Let

$$h(x) = \begin{cases} w(x) - w(0), & 0 \leq x \leq D, \\ -w(0), & x > D. \end{cases}$$

Using the Gauss integral for the digamma function, combining all κ_m shifts, and only then adding the endpoint W_0 term gives the exact identity

$$\sum_m \kappa_m E_2(s + dm) = -\eta \int_0^\infty \frac{e^{-\sigma y} p(e^{-dy}) h(\eta y) \cos(Ty)}{e^{2y} - 1} dy. \quad (69)$$

The central digamma integral and the endpoint term have therefore combined into one Bose–Einstein kernel.

Since $h(0) = h'(0) = 0$ and the certified Taylor bounds give

$$0 \leq -h(x) \leq \frac{51.74}{2}x^2 + \frac{144}{6}x^3,$$

Equation (69) yields an $O(\eta^3)$ error rather than an $O(\eta^2 \log t)$ error. Define

$$A_j(\sigma, d) = \int_0^\infty \frac{e^{-\sigma y} p(e^{-dy}) y^j}{e^{2y} - 1} dy.$$

Then

$$\left| \sum_m \kappa_m E_2(s + dm) \right| \leq \eta^3 \frac{51.74}{2} A_2 + \eta^4 \frac{144}{6} A_3. \quad (70)$$

Moreover, along $d = 2\sigma - 1$,

$$\frac{d}{d\sigma} \left(e^{-\sigma y} p(e^{-dy}) \right) = -2y e^{-\sigma y} q(e^{-dy}) < 0,$$

so the moments are maximized at the minimal allowed σ . Expanding the Bose factor gives the finite Hurwitz-zeta representation

$$A_j = \frac{j!}{2^{j+1}} \sum_m \kappa_m \zeta \left(j + 1, 1 + \frac{\sigma + dm}{2} \right), \quad (71)$$

which is enclosed by finite partial sums and elementary integral tails.

The exact gamma and combined-error ledgers give

$$4.77357 \longrightarrow 4.773125 \longrightarrow 4.7711 \longrightarrow 4.77085 \longrightarrow 4.77080 \longrightarrow 4.77079.$$

At this stage the explicit-formula error is no longer the dominant loss.

8.6 The complete weighted prime-power trajectory

The earlier arithmetic argument replaced the smoothing factor by a uniform constant. The later computation keeps the complete normalized weight. For $u = m \log p$, define

$$\omega_u(A, B, L) = e^{-\sigma(A, L)u} w \left(\frac{Bu}{L} \right) p \left(e^{-d(A, L)u} \right), \quad (72)$$

where

$$\sigma(A, L) = 1 - \frac{A}{L + \log(K + T_0 e^{-L})}, \quad d(A, L) = 2\sigma(A, L) - 1.$$

The identity

$$\frac{\partial}{\partial A} \left(e^{-\sigma u} p(e^{-du}) \right) = \frac{2u}{L + \log(K + T_0 e^{-L})} e^{-\sigma u} q(e^{-du}) > 0$$

and the monotonicity of w reduce the full parameter box to finitely many one-dimensional L -problems. For 25 primes and 15 powers there are 375 such weights. Directed dyadic subdivision certifies every minimum. After rational rounding, each prime orbit becomes an integer polynomial of degree at most 180 and is certified by exact Bernstein subdivision.

The resulting normalized arithmetic coefficient satisfies

$$\boxed{\sum_{p < 100} (\log p) \min_x \sum_{m=1}^{15} \omega_{m \log p}(A, B, L) P_{12}^{(2)}(mx) > 1.34064.} \quad (73)$$

The previous uniform coefficient was approximately 0.70912 in the same normalization. Thus the arithmetic sublemma nearly doubles, although the full zero-free constant changes much less because the denominator, gamma channels, and trajectory main term also change. Inserting (73) gives the finite chain

$$4.77079 \rightarrow 4.765 \rightarrow 4.763 \rightarrow 4.7625 \rightarrow 4.7622 \rightarrow 4.76217 \rightarrow 4.76216 \rightarrow 4.76215.$$

8.7 A degree-eight SOS factor

The degree-six discrete factor is close to the logistic limit $1/(1+x)$, but a degree-eight SOS construction captures most of the remaining finite-degree gain. Put

$$G(Y) = \left(Y - \frac{2}{5}\right) \left(Y - \frac{25}{4}\right) \left(Y - \frac{214}{5}\right) (Y - 577), \quad \varepsilon_8 = 10^{-10}.$$

For $p_8(x) = \sum_{m=0}^8 \kappa_m^{(8)} x^m$, write $(1+x)p_8(x) = \sum_{m=0}^9 b_m x^m$ and define

$$\mathcal{N}_\kappa(Y) = \sum_{m=1}^9 m b_m \prod_{\substack{1 \leq j \leq 9 \\ j \neq m}} (Y + j^2). \quad (74)$$

The exact rational vector $\kappa^{(8)}$ is the unique solution with $\kappa_0^{(8)} = 1$ of

$$\mathcal{N}_{\kappa^{(8)}}(Y) = c_8 (G(Y)^2 + \varepsilon_8 (1+Y)^8), \quad c_8 > 0. \quad (75)$$

Hence the strip-boundary Fourier density is strictly positive on the entire half-line. Numerically,

$$(\kappa_0^{(8)}, \dots, \kappa_8^{(8)}) \approx (1, -0.9999689162, 0.9989391882, -0.9875947320, \\ 0.9279060573, -0.7551425553, 0.4644648599, -0.1810107989, 0.0324082804),$$

with

$$\sum_m \kappa_m^{(8)} = 0.50000138347 \dots$$

Exact Bernstein and coefficient-sign certificates prove the analogues of all positivity and monotonicity statements used above. The fully weighted orbit lower bound is

$$> 1.3406610092 \dots,$$

and the resulting chain is

$$4.76215 \rightarrow 4.76190 \rightarrow 4.76185 \rightarrow 4.76182 \rightarrow 4.7618105.$$

Higher-degree diagnostics converge rapidly to the logistic factor $1/(1+x)$ and improve the formal wall by only about $1.3 \cdot 10^{-5}$ beyond this point.

8.8 Optimizing the Heath–Brown parameter

The main smoothing efficiency can be written as a Rayleigh quotient. If $w = g * \tilde{g}$ on $[-b, b]$, then

$$w(0) = \langle g, g \rangle, \quad W(0) = \frac{1}{2} \left(\int g \right)^2, \quad W(-1) = \frac{1}{2} \iint e^{|x-y|} g(x) g(y) \, dx \, dy.$$

The leading operator is therefore self-adjoint. In the one-parameter Heath–Brown family its stationary equation is

$$\sin^2 \theta = a_1^{(2)} (1 - \theta \cot \theta). \quad (76)$$

The leading-order solution is about 1.13864, while the complete error ledger moves the optimum slightly left. The rational choice

$$\theta = \frac{56859}{50000} = 1.13718 \quad (77)$$

is re-certified from scratch. The 375 weight problems and all 25 orbit polynomials remain positive, with

$$\mathcal{A}_{<100} > 1.3531341853. \quad (78)$$

The finite step

$$4.7618105 \longrightarrow 4.76180$$

has interval margin greater than $1.5 \cdot 10^{-7}$.

8.9 The large-prime third-harmonic tail

The certificate (78) only uses $p < 100$. For larger primes the first three harmonics already force a uniform positive gap. The exact trigonometric inequality

$$\boxed{9480P_{12}^{(2)}(x) + 97P_{12}^{(2)}(2x) + P_{12}^{(2)}(3x) \geq \frac{33}{50}} \quad (79)$$

was converted into a degree-36 integer polynomial and certified by exact Bernstein subdivision. In the final parameter box, the first three prime-power weights satisfy

$$\frac{W_1(p)}{W_3(p)} > 9480, \quad \frac{W_2(p)}{W_3(p)} > 97 \quad (101 \leq p \leq 997).$$

Thus every such prime contributes at least $0.66W_3(p)$. A refined piecewise version of (79), with twelve prime intervals up to 10^4 , yields the strict tail lower bound

$$\boxed{\sum_{101 \leq p \leq 10000} (\log p) \min_x \sum_{m \geq 1} W_m(p) P_{12}^{(2)}(mx) > 0.0001887564129717862.} \quad (80)$$

Combining the $p < 100$ and large-prime certificates permits the safe arithmetic coefficient

$$\mathcal{A}_{\text{final}} = 1.35331.$$

The final finite step is

$$\boxed{4.76180 \longrightarrow 4.761794,} \quad (81)$$

with directed-interval margin greater than $3.2 \cdot 10^{-8}$. A target 4.7617935 also passes, but with a margin below $5 \cdot 10^{-9}$ and is not used in Theorem 8.1.

8.10 Consolidated finite chain and numerical comparison

The complete post-4.8038 chain is summarized in Table 4. Some rows abbreviate several rationalized microsteps; every abbreviated block consists only of applications of (26) with an already certified seed.

module	endpoint	principal certified change
sharper signed-shift norm	4.80347	$72 \rightarrow 63.816$ in the Fourier L^1 bound
positive-definite boundary	4.77775	removal of the c_0 -Sturm wall
exact trajectory	4.77357	no rectangular Taylor loss in Lemma 14
exact gamma and Bose collapse	4.77079	combined central and endpoint E_2 error
fully weighted arithmetic	4.76215	trajectory weight retained for all 375 scales
degree-eight SOS	4.7618105	global half-line positivity with eight shifts
optimized θ	4.76180	$\theta = 1.13718$ and re-certified orbits
large-prime tail	4.761794	third-harmonic rigidity for $p \geq 101$

Table 4: Structural stages from the 4.8038 candidate to 4.761794.

Numerically,

$$\frac{4.8038 - 4.761794}{4.8038} \cdot 100 = 0.87443\% \quad \text{in } R,$$

while the width $1/R$ increases by

$$\left(\frac{4.8038}{4.761794} - 1 \right) \cdot 100 = 0.88215\%.$$

Relative to the original 4.8913 baseline, the total changes are 2.64768% in R and 2.71969% in the width.

8.11 Reproducibility obligations for the second supplement

A complete independent replay should verify the following modules separately.

Exact algebra. Verify the positive coefficient polynomial (64), the SOS identity (75), all Bernstein signs for $p, q, L, -L'$ and $1 + p'$, the coefficient signs proving the monotonicity of the auxiliary rational sums, and the harmonic inequalities (79) and their piecewise refinements.

Exact finite orbit arithmetic. For every certified prime range, rationally enclose the trajectory weights, form the integer Chebyshev orbit polynomial, and replay the dyadic Bernstein subdivision. The logarithms of primes should be enclosed by rational atanh series, not by binary floating-point constants.

Directed intervals. Replay the 63.816 Fourier bound, the exact-trajectory derivatives, the digamma and Bose moments, the 375 trajectory weights, all Ford remainders, and every finite-step margin. The final step (81) should be checked in an independent interval library because its margin is only of order 10^{-8} .

External inputs. Audit the precise hypotheses and height ranges of the classical seed 4.862, the Littlewood constant 19.62, the verified-zero computation, and the smoothed explicit formula imported from [1, 7].

9 Comparison of the three stages

The first supplement changed 4.8913 to 4.8038. The second changes 4.8038 to 4.761794. Across the complete manuscript the numerical change is

$$\frac{4.8913 - 4.761794}{4.8913} \cdot 100 = 2.64768\% \quad \text{in } R,$$

and

$$\left(\frac{4.8913}{4.761794} - 1 \right) \cdot 100 = 2.71969\% \quad \text{in the width } 1/R.$$

The gain does not come from one parameter. It decomposes as follows.

- (i) The baseline 4.8913 manuscript supplies the certificate architecture, the first Q_{12} , the initial prime-power grouping, and the corrected Laplace-transform ledger.
- (ii) The stronger external constants 4.862 and 19.62 permit a wider, but still non-circular, finite bootstrap.
- (iii) The second Fejér square and direct finite orbit certification increase the raw arithmetic floor to 0.2439742.
- (iv) Reoptimized degree-six κ vectors and actual Ford constants give the pre-Fourier candidate 4.8062105.

- (v) The signed-shift Fourier collapse eliminates a substantial loss caused by taking seven absolute values separately and gives 4.8038.
- (vi) Positive-definite boundary pairing, exact trajectory and digamma identities, fully weighted arithmetic, a degree-eight SOS factor, θ -optimization, and the large-prime third-harmonic tail give 4.761794.

The central methodological principle is the same one that appears in Selberg’s positive-source argument: combine the algebraically related terms before applying the triangle inequality. In the present setting the combination is not a Dirichlet convolution but the Fourier identity (36).

10 Updated reproducibility protocol

The supplemented computation separates into four layers.

Exact integer and rational arithmetic

1. Verify the norm and autocorrelations of (27).
2. For every $p < 100$, form the finite degree- $12M_p$ orbit polynomial and certify the chosen b_p by exact Bernstein subdivision.
3. For every bootstrap vector κ , certify Bernstein positivity of p and q , Sturm positivity of the Lemma 5 numerator, and coefficientwise signs for the derivatives of T_1 and T_2 .
4. Evaluate the moments (43)–(46) as exact finite rational sums.

Directed interval arithmetic

1. Enclose $w''(0)$ and $\sup |w'''|$, then prove (41)–(42).
2. Replay the $C = 72$ Carlson bound for every step vector.
3. Check the smoothing-function lower bounds, Ford remainders, gamma terms, and finite-step monotonicity.
4. Verify every margin in Tables 2 and 3.

External analytic inputs

The replay imports the smoothed explicit formula, zero-counting estimates, Ford transform bounds, the verified-zero height, and the two Yang zero-free constants as reported in [1, 7]. These are not reproved here.

Still missing

A publication-quality result requires an independent interval engine, a direct audit of the precise hypotheses behind the 4.862 and 19.62 inputs, a continuous rewrite of the modified BTY lemmas, and third-party review.

Second-supplement replay. The later stages additionally require the positive-definite boundary numerator, the exact trajectory and Bose-kernel identities, the degree-eight SOS linear system, the fully weighted prime-power trajectory certificates, the optimized- θ orbit replay, and the large-prime harmonic tail. The strongest reported endpoint 4.761794 should not be quoted without this complete dependency chain.

11 Concluding assessment

The two supplements produce a materially stronger internal candidate than the 4.8913 baseline, but the result remains an optimization of a classical zero-free-region argument. It does not produce a fixed half-plane, does not alter the asymptotic shape $1 - c/\log t$, and does not constitute a scale-level advance toward the Riemann hypothesis.

References

- [1] C. Bellotti, T. Trudgian and A. Yang, *Zero-free regions inspired by work of Heath-Brown*, arXiv:2603.21490, 2026.
- [2] C. Bellotti, *Explicit bounds for the Riemann zeta function and a new zero-free region*, J. Math. Anal. Appl. **536** (2024), Paper No. 128249.
- [3] R. P. Brent, D. J. Platt and T. S. Trudgian, *Accurate estimation of sums over zeros of the Riemann zeta-function*, Math. Comp. **90** (2021), 2923–2935.
- [4] K. Ford, *Zero-free regions for the Riemann zeta function*, Number Theory for the Millennium, II, A K Peters, 2002, pp. 25–56; updated preprint arXiv:1910.08205.
- [5] D. R. Heath-Brown, *Zero-free regions for Dirichlet L -functions, and the least prime in an arithmetic progression*, Proc. London Math. Soc. **64** (1992), 265–338.
- [6] H. Kadiri, *Une région explicite sans zéros pour la fonction ζ de Riemann*, Acta Arith. **117** (2005), 303–339.
- [7] E. S. Lee, *Minimal zero-free regions for results on primes between consecutive perfect k th powers*, arXiv:2602.14340, 2026.
- [8] M. J. Mossinghoff, T. S. Trudgian and A. Yang, *Explicit zero-free regions for the Riemann zeta-function*, Res. Number Theory **10** (2024), Paper No. 11.
- [9] D. J. Platt and T. S. Trudgian, *The Riemann hypothesis is true up to $3 \cdot 10^{12}$* , Bull. Lond. Math. Soc. **53** (2021), 792–797.
- [10] A. Yang, *Explicit bounds on $\zeta(s)$ in the critical strip and a zero-free region*, J. Math. Anal. Appl. **534** (2024), 128124.